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SeaLabMOS Data Pipeline

The Atlas Scientific EC_EZO circuit's datasheet stipulates that "when you enable the salinity output of the EZO-EC circuit it mathematically converts conductivity to salinity using the Practical Salinity Scale (PSS-78)." [\[source\]](#). It does this, however, using a default temperature of 25C that resets on power-on. While it is possible in the firmware to send an updated temperature to the EZO-EC mid-deployment, we chose to disregard all data besides the raw electrical conductivity reported from the EZO-EC.

This is because even following calibration, the salinity (and if enabled the total dissolved solids and specific gravity) values reported by the EZO-EC are incorrect. While it may be possible to chase accurate values "out of the box" from the EZO-EC, we chose to entirely disregard this data. Instead, we opted to do the PSS-78 calculation ourselves from the electrical conductivity value (in reported by the EZO-EC

MIS_1 correction in the lab looks quadratic while MIS_2 in the field looked scalar. Given what we know so far, I propose a two pipeline data method before calculating SP_from_C, depending on how the system performs: 1) scalar gain like $C_{corrected} = EC_GAIN \times (ec / 1000)$ and 2) $C = ec / 1000$, then $C_{true} = A1 \times C + A2 \times C^2$

I've been using:

sal_fw = raw firmware salinity, trash
sal_recalc = recomputed salinity from raw EC value
sal_recal_g = scalar gain correction (ex. * .9908)
sal_recal_qua = quadratic gain correction (ex. above)

The default SeaLabMOS CTD's firmware reports three values from the EZO-EC: electrical conductivity (ec), salinity, and total dissolved solids. Only the first is used.

Compute practical salinity using the ec values from the Atlas Scientific EZO-EC circuit and the system's in-situ temperature via PSS-78 by utilizing GSW-Python

SP_from_C[\[source\]](#)

The returned SP value *should* closely track the RBR, but with the signature of a multiplicative conductivity calibration error intrinsic to the specific SeaLabMOS CTD system (different error per system)

Compute a least-squares fit of recomputed vs. reference salinity over the calibration points to find the EC_GAIN correction

Apply the EC_GAIN correction factor to the original ec data, then recompute using GSW SP_from_C for final salinity values